Healthcare Clinic Data Cleaning & Analysis

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Agenda

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- Findings- Department Workload
- Findings- Revenue by City
- Findings- Gender Distribution
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Introduction

The Problem

- The raw patient dataset (500 records, 23 fields) contained multiple data quality issues
- Missing values, duplicate entries, inconsistent formatting, and invalid records
- Unreliable data prevented accurate reporting on patient volume, revenue, and department performance

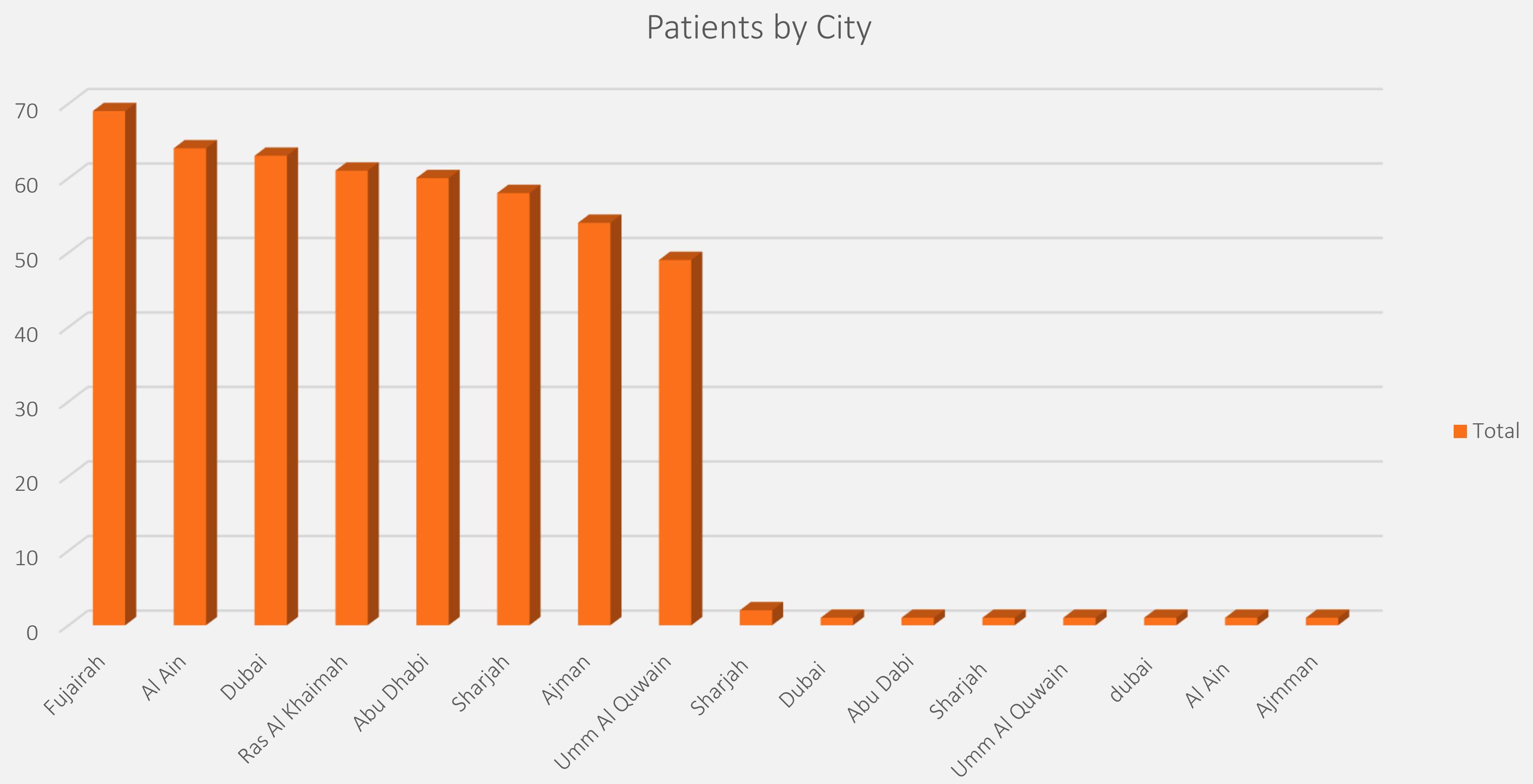
What I Did

- Cleaned and standardized 20+ fields (dates, phone numbers, emails, categorical values)
- Validated business logic (e.g., visit/discharge date sequencing)
- Identified and flagged duplicate and unresolvable records for manual review
- Documented every change in a Data Cleaning Log
- Built Pivot Tables and charts to extract business insights

Findings

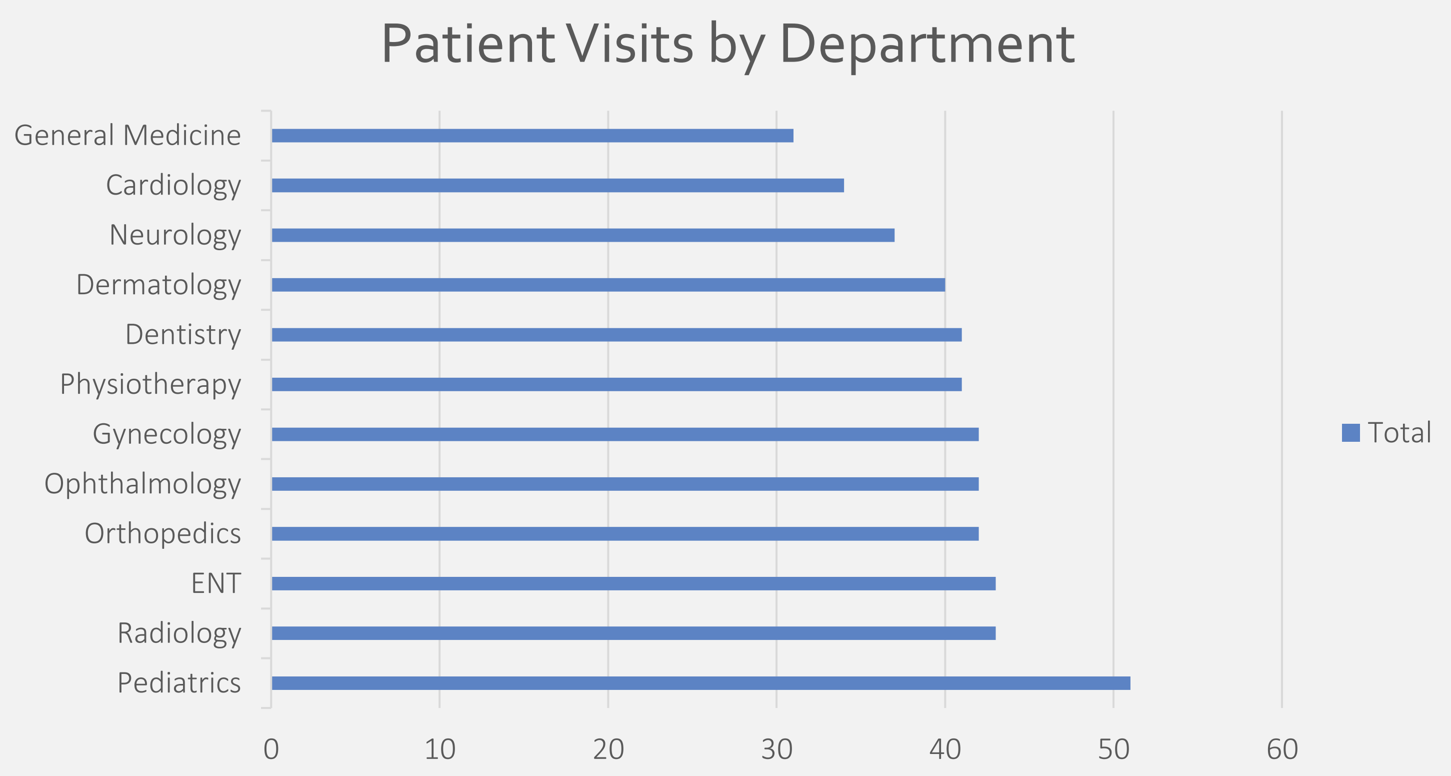
Let’s dive in

Where Are Our Patients?



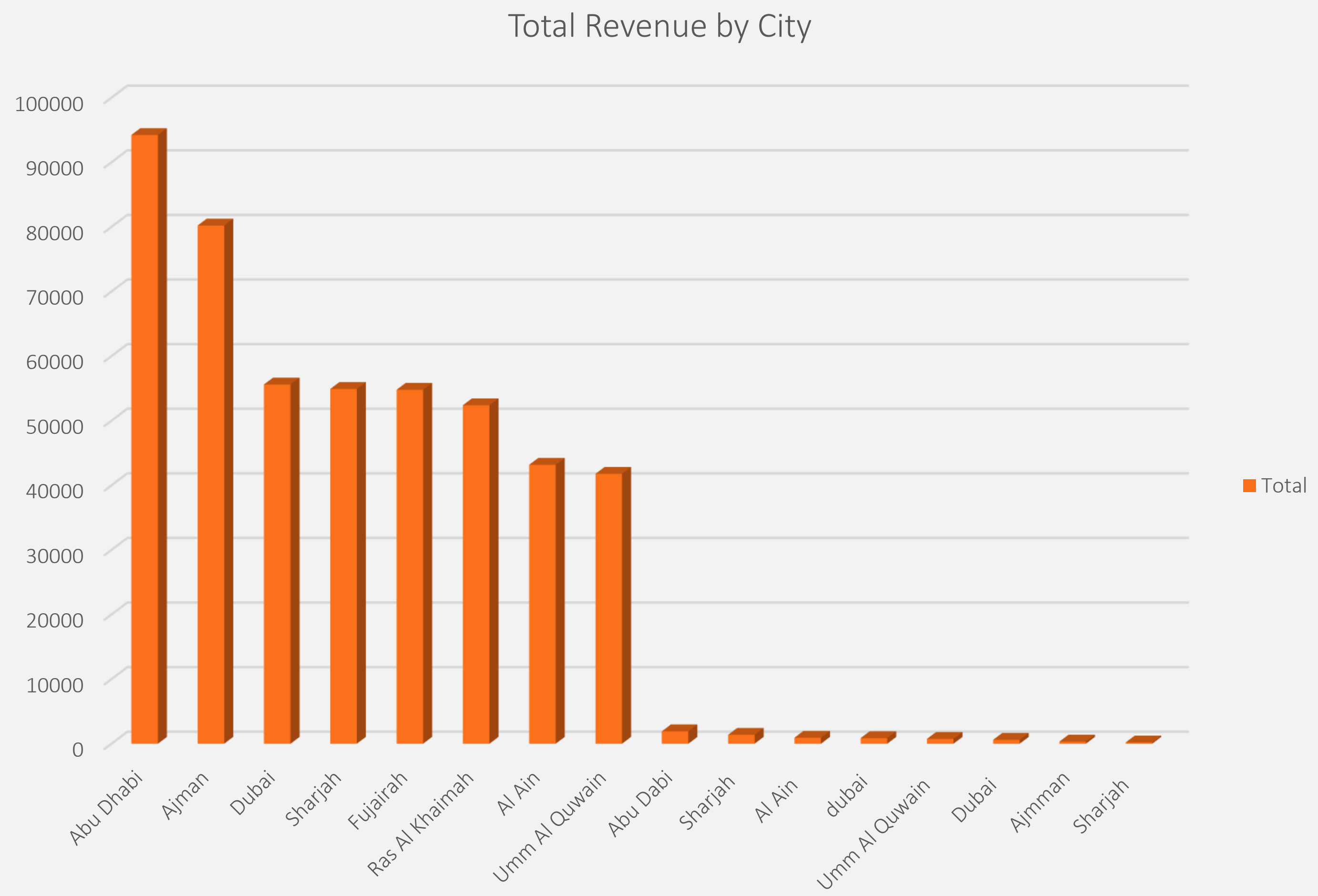
Fujairah has the highest number of registered patients, with Al Ain in second place.

Department Workload



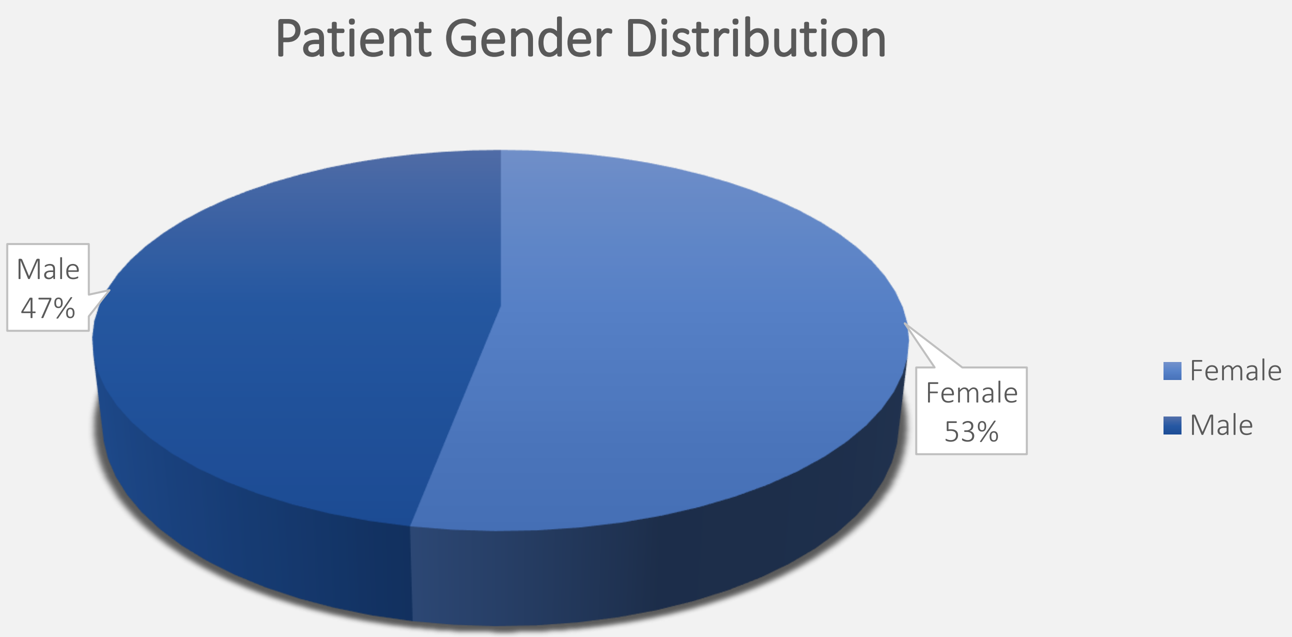
Paediatrics has the highest patient volume of any department at the clinic.

Which City generates the highest Revenue



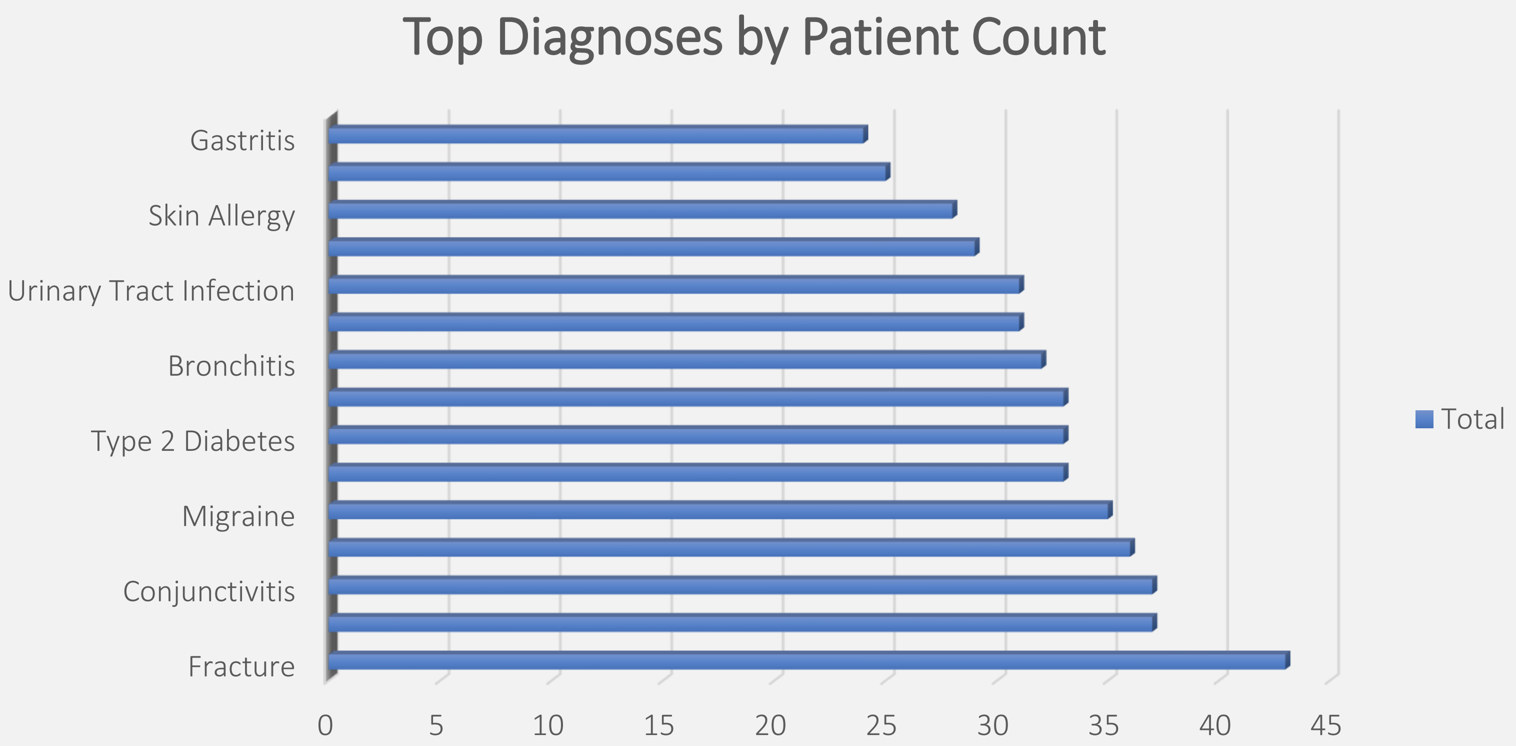
- Abu Dhabi generates the highest total revenue among all clinic locations. Notably, this doesn't match the city with the most patients (Fujairah), suggesting higher revenue per patient in Abu Dhabi.

Patient Demographics



The patient base is fairly balanced
53% female, 47% male.

Most Common Diagnoses



Fractures are the most common diagnosis (43 cases), followed by conjunctivitis (37 cases)

Data Quality Summary

Overall

- Dataset: 500 patient records, 23 fields
- 70% of records required some form of cleaning or correction

Issues Found & Resolved

- **Phone numbers**: Inconsistent formats standardized across the column (mixed country code prefixes, spacing, missing leading zeros) → 2 records with unrecoverable short numbers and 1 missing value flagged, not corrected
- **Email addresses**: Invalid formats corrected (missing "@", missing domain extension, stray spaces) → ~6 missing values identified, 3 confirmed duplicate emails linked to different patients flagged for manual verification
- **Date logic**: 4 records had a Registration Date occurring after the Visit Date; 5 records had a Discharge Date occurring before the Visit Date — all flagged as business logic errors requiring manual verification

Records Requiring Manual Follow-Up

- A total of 45 records could not be confidently auto-corrected and were compiled into a separate Follow-Up Log for review

Conclusion

Tools & Skills Applied

Microsoft Excel

Power Query (data transformation, formula-based validation)

Pivot Tables & Pivot Charts

Data quality auditing and documentation

Business insight reporting

Closing Statement

This project reflects a full data analysis workflow, from identifying and resolving data quality issues, to documenting decisions transparently, to translating a cleaned dataset into actionable business insights. Beyond the technical cleaning itself, a key focus was knowing when to correct data confidently versus when to flag it for human review a distinction that matters as much in real analyst work as the cleaning itself.